

Honours Thesis - Simulation of Squeezed Light Generation in an Optical Resonator

Aaron Dayton

CONTENTS

IProject Motivation	1
IIBackground Information	2
AOptical Non-Linearities	2
BElectromagnetically Induced Transparency	2
CSqueezed States	2
DThree-Level Atom	2
EFour Wave Mixing	2
FContinuous Variable Entanglement	2
GFock Basis	2
HMaster Equation	2
IIIThe first steps	2
IVHamiltonian Formulation	3
VComputational Methods	3
VIResults	3
VIIscussion	3
AConditions for Optical Squeezing	3
References	3

I. PROJECT MOTIVATION

MacRae's Outline:

Quantum fluctuations of the electromagnetic field impose a fundamental limit on measurement sensitivity, known as the standard quantum limit (SQL). Exotic quantum optical states known as "squeezed light" can surpass the SQL. Squeezed states are a hot topic in research, particularly in quantum communication, metrology, and optical quantum computing. This project will explore theoretical models predicting strong squeezing when atoms are placed inside an optical cavity. The student will develop and run numerical simulations to investigate the conditions for optimal squeezing, using tools from quantum optics and cavity QED theory. Through this work, the student will gain experience in programming, numerical methods, and the theoretical framework of quantum light-matter interactions.

The past hundred years since the formulation of quantum mechanics have led to unrivaled technological progress, especially in recent years. With greater understanding of and ability to control quantum phenomena further possibilities are becoming available. These include quantum key distribution (QKD), quantum communication, and quantum computation, to name a few.

Any emerging technology tends to follow the rule ascribed to increasing system complexity sometimes stated as 'more is different' with reference to the 1972 paper by Physicist Philip W. Anderson [1]. This especially applies to that of quantum networks. As greater quantum control is achieved, greater quantum networking capabilities will be required, thereby necessitating the development of improved methods for transmitting quantum information.

Currently, a hurdle in developing quantum networks is transmitting quantum information with a sufficiently long coherence time such that the information produced may actually be used. This is especially true if the information is to be transmitted across vast distances through optical fibres, such as China has done in a 4600 kilometer land-based network [2].

One dimension of this problem is that even if a coherent optical field is produced, it may not be at a frequency for which optical fibres have a low-loss window. Thus, effective frequency conversion methods which preserve quantum information must be developed.

This thesis includes a theoretically viable method for frequency conversion of two entangled electromagnetic fields (continuous-variable entanglement). The key idea is to send the entangled beams into an optical spherical resonator filled with cooled three-level atoms (Λ -system) such that a four wave mixing (FWM) process occurs to produce two entangled output fields. If the output frequencies are tuned such that they incur low-loss within an optical fibre, then the signals could theoretically be transmitted over long distances with existing technology. For a diagrammatic representation of this process, see figure 1. **TODO - Should squeezed states be mentioned here, or is it fine to leave them for later?**

The outline of this project can be described in three stages: formulating up the system Hamiltonian, computationally solving for energy eigenstates of the Hamiltonian, and analyzing the results to find how changing specific parameters may change the system behavior.

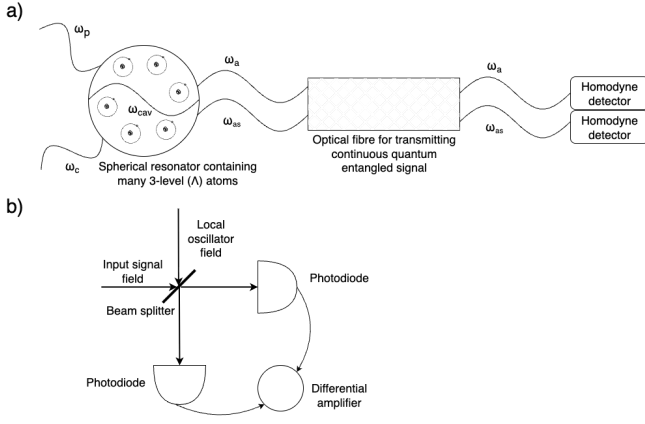


FIG. 1: Suggested setup of a real-world implementation for the four wave mixing process described with (a) two continuous beams entangled beams (continuous-variable entanglement) at frequencies ω_p and ω_c which undergo a four wave mixing process within a spherical resonator tuned to frequency ω_{cav} filled with cooled three-level atoms to produce Stokes and Anti-Stokes beams at frequencies ω_a and ω_{as} , respectively. The resulting beams are then may then be sent through an optical fibre to homodyne detectors. (b) Homodyne detector layout.

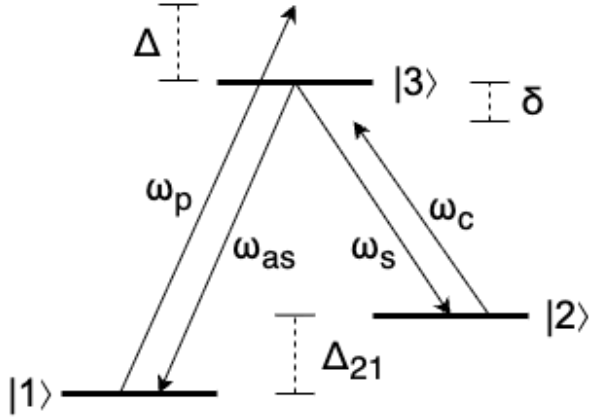


FIG. 2: **TODO**

II. BACKGROUND INFORMATION

A. Optical Non-Linearities

B. Electromagnetically Induced Transparency

C. Squeezed States

D. Three-Level Atom

E. Four Wave Mixing

F. Continuous Variable Entanglement

G. Fock Basis

H. Master Equation

III. THE FIRST STEPS

In order to work up to modelling a three-level atom in the Fock basis for photons with dissipative Lindbladian operators, a simpler model will be taken as a starting point. As is a common theme in physics, a hydrogenic atom in a time dependent electromagnetic field can be approximately modelled as a drive harmonic oscillator. So rather than dealing with quantized photons, the system will first be modelled using classical oscillating electromagnetic fields with frequencies nearly resonant with those of the atomic transitions in the three-level system.

To do this, a few assumptions will be made [?]: first, the wavelengths of the light sources interacting with the atom have wavelengths much greater than the Bohr radius of the atom ($\lambda \gg a_0$); second, the detunings of incoming beams are small in proportion to the transition frequencies considered. These two assumption happen to be consistent with one another.

The first assumption allows for us to approximate the electromagnetic field experienced by the atom as constant in space but varying in time since the change in electromagnetic field strength between the two ends the atom in the transverse wave direction is relatively small.

There exists the relation between the electric and magnetic field strengths of light which arises from Maxwell's equation [?]

$$B = \frac{E}{c}. \quad (1)$$

The Lorentz force is

$$\vec{F} = q \left(\vec{E} + \vec{v} \times \vec{B} \right) \quad (2)$$

where q is charge, and $\vec{v}$ is particle velocity. Using the Bohr model for approximation, we have that quantized electron velocities for a hydrogenic atom have magnitude

$$v_n = \alpha \frac{Z}{n} c \quad (3)$$

where α is the fine structure constant, Z is the atomic number, and n is the principle quantum number [?]. We can therefore see by rough approximation that we have

$$|\vec{v} \times \vec{B}| \lesssim \alpha Z c \frac{E}{c} = \alpha Z E. \quad (4)$$

We are considering Rubidium with atomic number $Z = 37$, but this neglects the effects of screening caused by the electrons in inner shells of the atom. Electron position is distributed in a cloud about the atom for any orbital, though we can expect the 36 electrons of inner shells between the valence electron (singular in its shell) and the nuclear charge of $37e$ to roughly cancel $36e$ of charge such that the valence electron experiences about $1e$. By this line of reasoning we have

$$|\vec{v} \times \vec{B}| \lesssim \alpha E. \quad (5)$$

Thus, the force due to the external magnetic field is about $1/137$ that of the external electric field. So we may neglect the external magnetic field for simplification.

$$\vec{F} \approx q\vec{E} \quad (6)$$

All of these approximations can be justified from the perspective that this model is a first estimate for system behavior which will be refined and iterated upon.

This brings us to a physicist's favorite abstraction: the harmonic oscillator. The potential of which the valence electron experiences can be expanded in a Taylor series and approximated as quadratic [?]. The mass of the nucleus is much greater than the electron mass, so we may approximate the nucleus to be fixed. So we have a mass with some equilibrium distance from a fixed point under a quadratic potential, i.e. a simple harmonic oscillator. We may then add in our external time-dependent driving force to obtain a driven harmonic oscillator.

When a charge oscillates, it emits photons as its movement produces a changing electric and magnetic field. So when our system is driven on resonance, then an electromagnetic field of that same frequency will be produced. Here lies the idea behind electromagnetically induced transparency. By driving a cloud of these hydrogenic

atoms on resonance, the field essentially ‘passes through’ the medium.

TODO - Connect these ideas.

The polarization vector of the atomic cloud may be given by

$$\vec{P} = N \langle \hat{d} \rangle = \epsilon_0 \chi \vec{E} \quad (7)$$

where N is atomic density, $\hat{d}$ is the average dipole moment, and χ is the magnetic susceptibility [?]. One can find that for a three-level atom (**TODO derive or cite?**)

$$\chi = \frac{-Nd^2}{\epsilon_0 \hbar} \frac{\delta}{|\Omega_c|^2 - \delta(\Delta + i\Gamma)} \quad (8)$$

where **TODO - Define detunings and dissipation rate along with a figure for a three-level atom.**

The Hamiltonian of the three-level atom can be given as

$$H_{atom} = \text{TODO}(\text{diagonal} - \text{slides}) \quad (9)$$

and the Hamiltonian of the light-matter interaction can be given as

$$H_{atom} = \text{TODO}(\text{see slides}) \quad (10)$$

The dissipative operators...

Solve master equation with...

Here's the plot...

IV. HAMILTONIAN FORMULATION

V. COMPUTATIONAL METHODS

VI. RESULTS

VII. DISCUSSION

A. Conditions for Optical Squeezing

[1] Philip W. Anderson, “More is different - broken symmetry and the nature of the hierarchical structure of science.” *Science* **177**, 393–396 (1972).

[2] Teng-Yun Chen et al. Yu-Ao Chen, Qiang Zhang, “An

integrated space-to-ground quantum communication network over 4,600 kilometres,” *Nature* **589**, 214–219 (2021).